List of essential things any game level editor should have

A great game level editor bridges the gap between creative design and technical execution. Whether built for internal developers or shipped to players, any essential level editor must include these core features:

1. WYSIWYG / Live Preview (What You See Is What You Get)

- **Real-time rendering:** The editor must display the world using the exact same engine, lighting, physics, and assets as the live game.
- **No blind tweaking:** Creators should never have to guess how a platform, particle effect, or light source will look by saving and launching a separate build. [1, 2]

2. Core Manipulation & Placement Tools

- **Transform gizmos:** Intuitive tools to move, rotate, and scale objects across axes (2D or 3D).
- **Grid and object snapping:** Options to align tiles, brushes, or props strictly to a grid or snap them flush against other surfaces to prevent gaps.
- **Free-cam navigation:** A fluid "flight" camera mode that lets designers zip around the map unconstrained by player collision or gravity. [1, 2, 3]

3. Layer and Hierarchy Management

- **Layer separation:** Clear control over background, foreground, terrain, collision, and trigger layers so assets don't overlap or block gameplay improperly.
- **Scene tree / Hierarchy panel:** A structured list of every object, entity, and light source in the level for easy selection and parenting.
- **Inspector / Properties panel:** A way to click any object and tweak its custom variables (e.g., enemy health, door lock keys, spawn rates, or sound triggers). [1, 2, 3, 4, 5]

4. Workflow & Safety Nets

- **Undo / Redo system:** A reliable history stack so experimental changes can be cleanly rolled back.
- **Hotkeys and shortcuts:** Fast keyboard commands for common actions like copy-pasting, duplicating, hiding layers, and toggling preview modes.
- **Validation and error checking:** Automated checks that flag broken links, missing player start points, unassigned textures, or unreachable areas. [1, 2]

5. Saving, Exporting, and Sharing

- **Standardized data serialization:** Saving level layouts in a clean, version-control-friendly format (such as JSON or engine-native resource files).
- **Playtest toggle:** A one-button shortcut to immediately drop the player character into the level to test jumps, combat, or pacing.
- **Community integration (if player-facing):** Built-in import/export tools or workshop integration to let players share creations online (such as Steam Workshop). [1, 2, 3, 4, 5]

6. Asset & Resource Browser

- **Centralized asset library:** A dedicated panel to search, filter, drag, and drop meshes, sprites, audio files, textures, and blueprints directly into the scene.
- **Smart tagging and collections:** The ability to group assets by biome, theme, or mechanical function (e.g., "Industrial Pack," "Destructible Props") to prevent designers from digging through deep folder structures.
- **Thumbnail generation:** Automatic visual previews for all 2D and 3D assets so creators can identify objects at a glance.

7. Global Level Configuration & Environment Controls

- **World settings panel:** A master control area to define level-wide rules, such as gravity multipliers, default weather, time-of-day cycles, and background music.
- **Post-processing profiles:** Quick toggles or sliders to test color grading, fog density, bloom, and ambient exposure to set the map's mood instantly.
- **Player baseline setups:** Structural definitions for player spawn points, starting inventory, and winning or losing criteria (e.g., trigger zones for "Kill All Enemies" or "Reach the Goal").

8. System Diagnostics & Performance Profiling

- **Live heatmaps or metrics:** Built-in counters for frame rates (FPS), draw calls, vertex counts, and memory usage within the editor window.
- **Budgets and limits indicators:** Visual warning systems (like a progress bar or color coding) that alert the designer if they have placed too many lights, physics objects, or high-poly models that will tank performance on target platforms.
- **Collision visualizers:** Toggles to show or hide invisible boundary walls, navmeshes, and triggers so designers can debug pathfinding and player clipping issues without guessing.

9. Extensibility, Customization, & UI Layout Flex

- **Dockable window layouts:** A modular interface where panels (Inspector, Asset Browser, Scene Tree) can be detached, resized, or moved to secondary monitors.
- **Custom tool scripting:** An open API or macro system allowing developers to build custom layout scripts (e.g., a "scatter grass" brush or an automatic wall-generator).
- **Workspace presets:** Quick layout switches optimized for different tasks, such as a "Lighting Layout," a "Blocking-out Layout," or an "AI Scripting Layout."

10. Multi-User Collaboration & Pipeline Safety

- **Asset locking / Source control hooks:** Structural integration with Git, Perforce, or SVN that shows who is currently editing a specific sub-level or asset to prevent overwrite conflicts.
- **Auto-save and crash recovery:** Background saving and local crash logs that ensure hours of precise placement work aren't lost if the editor freezes.
- **Sub-level / Chunk streaming support:** The architectural ability to split a massive level into smaller, modular chunks or layers so multiple designers can work on the same environment simultaneously.

Here are the essential QoL features, functions, and layouts that elevate a level editor from functional to professional:

1. High-Efficiency Placement Functions

- **Object Scattering & Brush Painting:** Instead of placing 500 rocks individually, a brush tool that lets designers "paint" foliage or props onto the terrain with randomized scale, rotation, and density.
- **Replace in Selection:** The ability to highlight 20 old streetlamp assets in a level and swap them out for a new model with a single click, keeping all original spatial positions intact.
- **Prefab / Blueprint Creation:** Grouping a collection of individual objects (e.g., a table, chairs, plates, and a candle) into a reusable template that can be stamped down instantly across multiple maps.
- **Drop to Floor:** A dedicated hotkey that instantly snaps floating props down to the nearest collider beneath them, eliminating manual vertical tweaking.

2. Micro-Navigation & Contextual Layouts

- **Contextual Right-Click Menus:** Right-clicking an object should instantly surface the most relevant actions for *that specific object* (e.g., right-clicking a light brings up "Isolate Light," right-clicking an enemy brings up "Open AI Behavior Tree").
- **Bookmark Views:** The ability to save camera positions (e.g., Ctrl + F1 saves the Boss Room view), allowing designers to jump instantly across massive maps without flying or zooming manually.
- **Isolate Selection (Focus Mode):** A hotkey to temporarily hide everything in the world except the single object or building the designer is currently editing, removing background clutter.
- **Breadcrumb Navigation:** A clear path bar at the top of the viewport showing exactly where you are inside nested object groups or sub-levels (e.g., `World > Castle Zone > Keep Interior > Throne Asset Group`).

3. Smart Asset Management QoL

- **Favorite Assets Bin:** A persistent, customizable tray where designers can pin their 10 most frequently used props, tiles, or triggers for instant access.
- **Drag-and-Drop Replacements:** Holding a modifier key (like Alt) while dragging an asset from the browser directly onto a placed object to instantly replace it.
- **Broken Reference Auto-Fixers:** If an asset path changes, a prompt that asks, *"Asset missing. Do you want to re-route all references to the new folder location?"* instead of throwing individual errors.
- **Color-Coded Folders & Entities:** The ability to assign custom colors to scene folders or hierarchy items (e.g., red for enemy spawns, blue for audio triggers) for quick visual scanning.

4. Ideal Workspace UI Layout (The "Golden Standard")

A highly efficient editor layout should be modular, but its default state typically organizes information to match a designer's muscle memory:

Layout Zone	Primary Purpose	QoL Enhancements
Central Viewport	3D/2D Level Canvas	Maximized workspace with floating, minimal overlays for grid settings and camera speed. Supports full-screen mode toggle (`F11`).

Left Sidebar	Hierarchy / Scene Tree	Includes a quick filter search bar. Shows visibility icons (eye tool) and edit-lock icons right next to every object.
Right Sidebar	Inspector / Property Panel	Collapsible sections. Number inputs support clicking and dragging to slider-adjust values, or math inputs (e.g., typing `+5` into the X-coordinate).
Bottom Dock	Asset Browser & Console Log	Multi-tab support so you can keep your assets pack open next to the real-time bug/error console. Collapses out of sight when not active.
Top Menu Bar	Transform Tools & Testing	Quick-toggle icons for snapping increments, local vs. world coordinate spaces, and a massive, unmissable **"Play Level"** button.